

Role Constellations in Sustainability Transitions

Draft Chapter for Review

RoleBox Dissertation

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Chapter Overview

This chapter develops the theoretical framework for understanding **role constellations**—relational configurations among actor roles that shape sustainability transition dynamics. We situate role constellations within the Multi-Level Perspective and Multi-Actor Perspective, identify research gaps in current scholarship, and introduce the EIT Knowledge and Innovation Communities as an empirical lens for exploring cross-domain constellation patterns.

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1 Conceptual Foundations

Sustainability transitions research has historically prioritized structural-systemic analyses, with the Multi-Level Perspective (MLP) serving as the dominant framework. However, scholars note that transition research “lacks a suitable vocabulary to analyse the (changing) interactions and relations of actors.” While agency is acknowledged as integral to transitions, the operationalization of actor interactions remains limited.

Role constellations refer to relational configurations among actor roles—patterns of cooperation, competition, and mutual influence shaping transition dynamics. We define *roles* as recognizable sets of activities addressing recurring situations, and *transition roles* as those through which actors support or hinder specific transitions. Related terminology includes *actor constellations*, *actor configurations*, and *practice constellations*.

2 Theoretical Frameworks

The Multi-Actor Perspective (MaP) distinguishes four institutional sectors (state, market, third sector, community) and three aggregation levels (sectors, organizations, individuals), with a hybrid sphere of boundary-crossing intermediaries. Wittmayer and colleagues operationalized sociological role theory for transitions, identifying constellation dynamics including cooperative, antagonistic, and independent relationships.

Bodenheimer linked roles to transition phases via the Dialectic Issue Lifecycle Model, showing how constellations evolve from problem emergence (issue identifiers vs. status quo maintainers) through political debate (coalition builders, framers) to stabilization (institutional integrators). Key typologies distinguish niche actors (innovation champions, frontrunners) from regime actors, now recognized as heterogeneous. Intermediary typologies identify systemic, regime-based, niche, process, and user intermediaries.

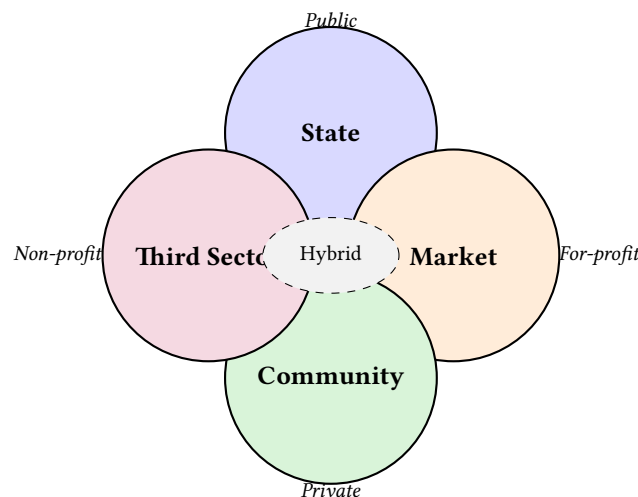


Figure 1: Multi-Actor Perspective: Four institutional sectors

3 Research Gaps

Despite significant field growth, explicit constellation theorization remains underdeveloped. Most studies capture static configurations rather than tracking evolution across phases. Integration across micro (psychology), meso (organizations), and macro (institutions) levels is limited. Non-human agency is largely neglected despite material-discursive turns. Empirically, Global South contexts remain un-

derrepresented, energy transitions dominate sectoral coverage, and multi-system interactions require further investigation.

Table 1: Research gaps in role constellation scholarship

Gap Type	Description
Weak theorization	Constellations invoked but rarely operationalized
Static analysis	Snapshots dominate; evolution understudied
Cross-level integration	Individual–organizational–systemic levels disconnected
Geographic bias	Global South underrepresented
Sectoral imbalance	Energy dominates; agri-food, buildings neglected
Incumbent heterogeneity	Limited differentiation among incumbent types
Multi-system dynamics	Cross-sector interactions poorly understood

4 Research Opportunities

We identify seven priority directions for advancing role constellation research:

1. **Systematic typology development:** Classify constellation patterns (cooperative, competitive, hybrid, emergent) beyond ad hoc descriptions.
2. **Longitudinal studies:** Track constellation evolution across transition phases, identifying tipping points and reconfiguration moments.
3. **Multi-scale integration:** Bridge psychological micro-foundations with organizational dynamics and institutional conditioning.
4. **Incumbent peripheral actors:** Investigate how actors from adjacent regimes (banks, universities, regulators) shape transitions.
5. **Comparative cross-context analysis:** Systematic comparison across sectors, regions, and governance scales.
6. **Methodological pluralism:** Combine network analysis, computational text methods, and participatory approaches.
7. **Governance applications:** Develop actionable constellation diagnostics for policymakers and transition managers.

A “constellation turn” complementing the established niche-regime-landscape focus with systematic attention to relational configurations could significantly advance both theoretical understanding and practical governance of sustainability transitions.

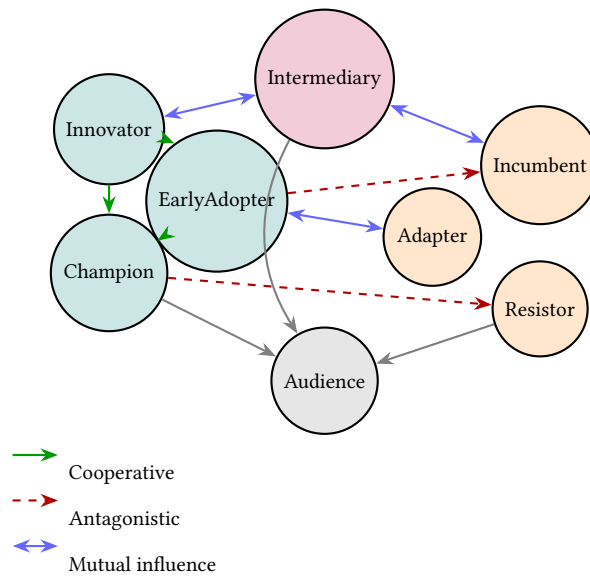


Figure 2: Role constellation dynamics: cooperation, antagonism, and influence

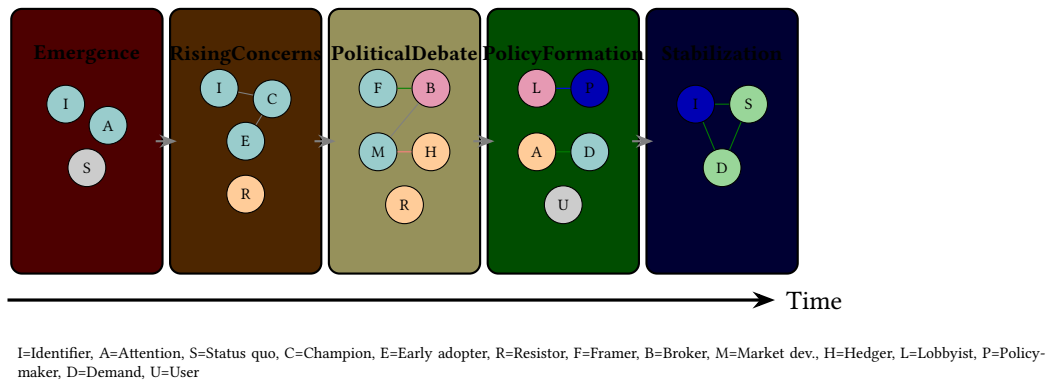


Figure 3: Role constellation evolution across transition phases

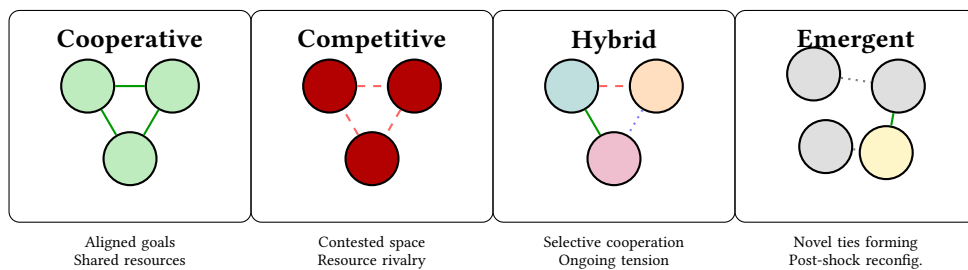


Figure 4: Typology of role constellation patterns

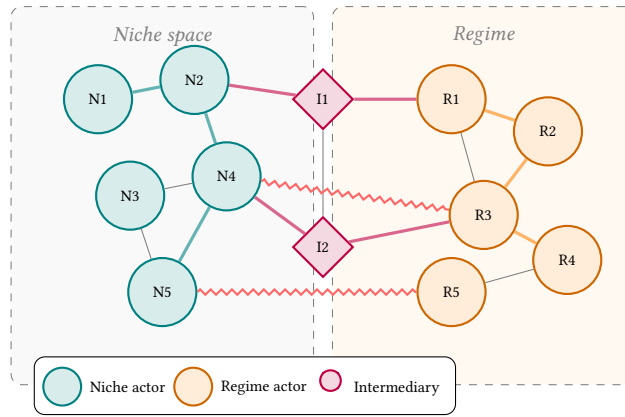


Figure 5: Niche-regime constellation network with intermediary bridging

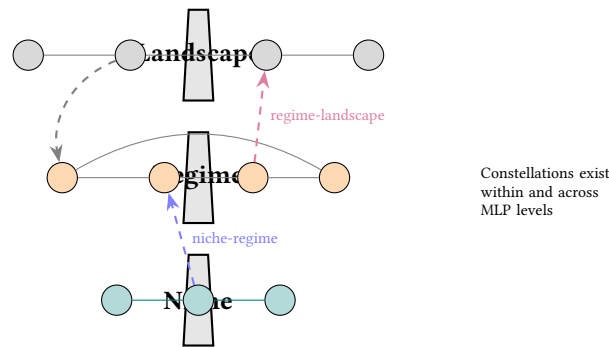


Figure 6: Multi-level view: constellations within and across niche, regime, landscape

5 Role Constellation Dynamics

6 Phase-Constellation Evolution

7 Constellation Typology

8 Niche-Regime Constellation Networks

9 Multi-Level Constellation View

10 Looking Across: The Eight EIT Intermediaries

The EIT comprises eight Knowledge and Innovation Communities, each an intermediary anchoring its own constellation:

- Climate-KIC (2010), Digital (2010), InnoEnergy (2010)
- Health (2015), Raw Materials (2016), Food (2017)
- Manufacturing (2019), Urban Mobility (2019)

Rather than assuming we know what patterns exist, we pose open questions: *What do we see when we look across these intermediaries? What modalities help us perceive imbrications—the overlapping, interlocking patterns across domains?*

10.1 The Eight Intermediaries as Probes

Each KIC is a probe into a different sustainability challenge. Laying them side by side invites comparative seeing.

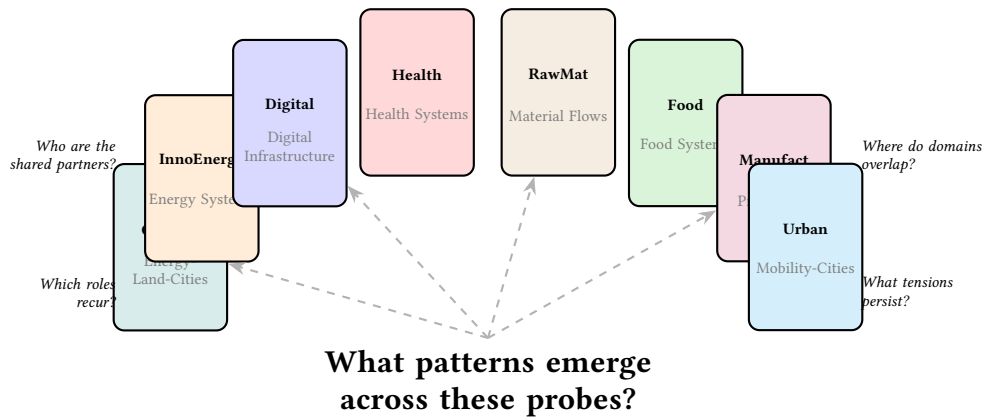


Figure 7: Eight EIT KICs as probes into different sustainability domains

10.2 Modality 1: Partner Overlap Matrix

One way to see imbrication is through shared membership. Which organisations participate in multiple KICs? The overlap matrix reveals cross-domain bridging actors—organisations whose participation stitches constellations together.

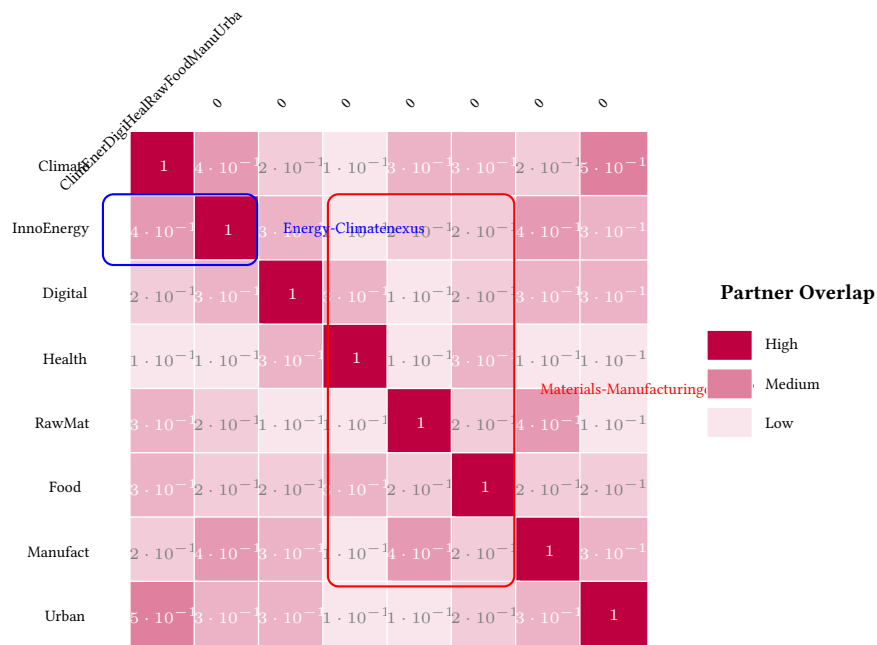


Figure 8: Partner overlap matrix across eight KICs (illustrative)

11 From Connections to Roles: The EIT Vignette

Understanding role constellations requires beginning where transitions actually unfold—at the level of concrete actor connections. This vignette examines the EIT Knowledge and Innovation Communities

through a bottom-up lens, tracing how local relational patterns aggregate into recognizable constellation structures.

11.1 Starting at the Ground: Local Actor Connections

Consider a Deep Demonstration site—Leuven, Belgium, one of 14 cities in EIT Climate-KIC’s Healthy, Clean Cities programme. At this granular level, we observe discrete dyadic and triadic connections:

- A **municipal climate officer** coordinates with a **university spin-off** developing building retrofit solutions
- A **local connector**—a practitioner embedded by design partner Democratic Society—facilitates dialogue between **citizen groups** and **city planners**
- A **regional utility** pilots smart grid technology with **SME suppliers**, mediated by **KIC project managers**

These connections are not random. Certain relational patterns recur:

Pattern 1: Brokered triads. Two actors lacking prior relationship connect through a third-party broker. The local connector links citizens to planners; the KIC links spin-offs to utilities.

Pattern 2: Knowledge-practice bridges. University researchers connect to municipal practitioners through joint pilot projects.

Pattern 3: Legitimacy chains. Start-ups gain credibility through association with established corporates, who in turn gain innovation credentials.

11.2 Emergent Roles from Connection Patterns

Roles are not assigned *a priori* but emerge from relational positioning:

Table 2: Emergent roles from observed connection patterns in Deep Demonstrations

Emergent Role	Connection Signature	Observed Actors
Orchestrator	High betweenness; bridges clusters	KIC portfolio managers
Local connector	Bridges citizen–government gap	Democratic Society practitioners
Knowledge producer	Outward ties to multiple adopters	Universities, RTOs
Niche carrier	Ties to orchestrator + early adopters	Start-ups, spin-offs
Anchor tenant	High degree within business cluster	Large corporates, utilities
Legitimacy broker	Certification ties across boundaries	KIC, city government
Demand articulator	Aggregates citizen input upward	NGOs, citizen platforms

12 Chapter Summary

Summary

This chapter developed the theoretical framework for **role constellations**—relational configurations among actor roles that shape sustainability transition dynamics.

Key theoretical elements:

- **Role constellations** are patterns of cooperation, competition, and mutual influence among

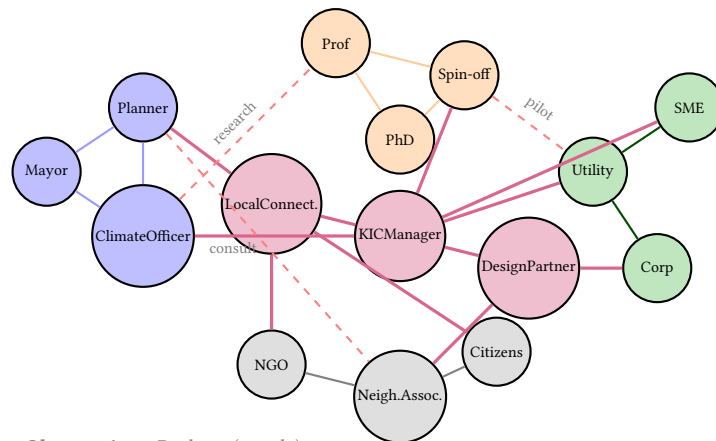


Figure 9: Ground-level actor connections in a Deep Demonstration site

actors occupying distinct transition roles

- The **Multi-Actor Perspective** provides the institutional mapping (state, market, third sector, community) with intermediaries in hybrid positions
- **Constellation dynamics** evolve across transition phases, from emergence through stabilization
- **Research gaps** include weak theorization, static analysis, cross-level integration, and geographic/sectoral bias
- The **eight EIT KICs** serve as probes into different sustainability domains, enabling comparative constellation analysis
- **Roles emerge from connections**—structural position in networks determines functional role, not formal designation

The next chapters apply this framework empirically, mapping role constellations across the EIT ecosystem using multimodal cartography.